

Reg. No:

Name:

APJ ABDULKALAM TECHNOLOGICAL UNIVERSITY
FIRST SEMESTER M.TECH. DEGREE EXAMINATION JANUARY 2025

Course Code: 221ECS004

Course Name: Computational Intelligence

Max. Marks: 60

Duration: 2.5 hours

Branch: Computer Science and Engineering

PART A

(Answer all questions in PART A. Each question carries 5 marks)

- | Q. No. | | Marks |
|--------|---|-------|
| 1 | <p>Let $U = \text{Animals} = \{\text{Dog, Cat, Horse, Cow, Lion, Tiger, Elephant}\}$ be a universe on which two fuzzy sets, one of Friendly animals and the other one of Strong animals, are defined as shown below.</p> <p>$P = \text{Friendly animals} = \{(\text{Dog}, 0.9), (\text{Cat}, 1.0), (\text{Horse}, 0.7), (\text{Cow}, 0.8), (\text{Lion}, 0.3), (\text{Tiger}, 0.2), (\text{Elephant}, 0.5)\}$</p> <p>$Q = \text{Strong animals} = \{(\text{Dog}, 0.4), (\text{Cat}, 0.2), (\text{Horse}, 0.9), (\text{Cow}, 0.6), (\text{Lion}, 1.0), (\text{Tiger}, 1.0), (\text{Elephant}, 0.8)\}$</p> <p>Compute the fuzzy sets $P \cup Q, P \cap Q, P', Q', P - Q, P \oplus Q$ Also verify that $P \cup P' \neq U$ and $P \cap P' \neq \emptyset$</p> | (5) |
| 2 | Explain the concept of Generalized Modus Tollens (GMT). How does it differ from GMP? | (5) |
| 3 | Briefly describe multi-objective optimization and pareto optimal set. | (5) |
| 4 | Explain the process of pheromone updating in Ant Colony Optimization. What are elitist and elastic ants, and how do they contribute to the solution? | (5) |
| 5 | Discuss the roles and significance of <i>pbest</i> and <i>gbest</i> particles in solving problems using Particle Swarm Optimization. | (5) |

PART B

(Answer any five full questions in PART B. Each full question carries 7 marks)

- 6 a. Briefly explain Crisp Sets vs. Fuzzy Sets, Operations on fuzzy sets, Linguistic Variables. (3)
- b. A goal of fuzzy logic is to achieve "computing with words". Briefly explain the idea. (4)

- 7 A Mamdani fuzzy inference system is used in a temperature control application with the following rules: (7)

- (i) IF Temperature is Cold THEN Fan Speed is Low.
(ii) IF Temperature is Warm THEN Fan Speed is Medium.
(iii) IF Temperature is Hot THEN Fan Speed is High.

The fuzzy sets are defined as:

Cold: $1.0/5 + 0.8/10$

Warm: $0.4/10 + 1.0/15 + 0.7/20$

Hot: $0.6/20 + 1.0/25 + 0.8/30$

Given the input temperature described by the fuzzy set (Temp = $0.5/15 + 0.9/20 + 0.4/25$), use the max-min compositional rule of inference to calculate the fuzzy output for the fan speed before defuzzification.

- 8 Initial population of a genetic algorithm implementation for solving a problem is shown in the table below. The outcome of applying the selection operator on the initial population is also shown. (7)

String	Fitness	$p(s_i)$ probability of selection of string = $f_i/\sum f$	Expected count = $f_i/\bar{f}$	Actual count (via roulette wheel selection)
S_1 : 1010	3	3/9	$3/(9/4) = 4/3$	1
S_2 : 0000	1	1/9	$1/(9/4) = 4/9$	0
S_3 : 1101	2	2/9	$2/(9/4) = 8/9$	1
S_4 : 0110	3	3/9	$3/(9/4) = 4/3$	2

- i. Give an explanation of the entries shown in the above table.
- ii. If one point crossover method is used, derive two possible populations for the next generation. Assume that the crossover and mutation probabilities are 0.9 and 0.1, respectively.

- 9 Demonstrate the first two rounds of applying the Particle Swarm Optimization (PSO) algorithm to minimize the function (7)
- $f(x,y) = (x + 3y - 6)^2 + (3x + y - 4)^2$. Use 4 particles with initial positions (6,2), (-2,8), (-8,3) and (-3,4). Set the inertia, cognitive, and social constants to 1.
- 10 How does each of the following three categories of algorithms attempt to balance (7)
between exploitation and exploration of the solution space:
- i. Genetic Algorithms
 - ii. Ant Colony Optimization Algorithms
 - iii. Particle Swarm Optimization Algorithms.
- 11 Explain the operation of the Ant Colony Optimization (ACO) metaheuristic by (7)
illustrating its application to the Travelling Salesman Problem (TSP) or any other combinatorial optimization problem.
- 12 Illustrate with an example the functioning of a fuzzy inference system. (7)